

Master's Presentation: Safety Concepts in Machine Learning

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MS Mathematical Sciences, Applied Mathematics

The following presentation is submitted to fulfill the project requirement of the University of Arkansas Little Rock degree of Master of Science in Mathematical Sciences, with emphasis in Applied Mathematics.

The material herein is drawn from 13 weeks of background reading, discussion with experts, and reflection.

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applications of machine learning

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Dr. Jones spent 2 to 4 hours a week listening to me complain and providing me with ideas.

Introduction to the Problem

Project Topic Evolution

Formally verified safety conditions in machine learning.

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Machine learning is very diverse.

Theory is not conducive to safety considerations.

What does it mean to be “safe”?

Project Topic Evolution

Formally verified safety conditions in machine learning.

Machine learning is very diverse.

Theory is not conducive to safety considerations.

What does it mean to be “safe”?

Identifying safety properties in machine learning.

Safety interpretation involving characterized behavior.

Broad framework for treating diverse problems.

What is Machine Learning

“As regards machines, we might say, very broadly, that a machine learns whenever it changes its structure, program, or data(based on its inputs or in response to external information) in such a manner that its expected future performance improves.”

–Introduction to machine learning: An early draft of a proposed textbook, Nils Nilsson, Stanford, 2005

What is Machine Learning

Broadly speaking, two types:

- Batch Learning

- Iterative (Online) Learning

Iterative learning poses interesting problems, but the static nature of batch learning makes it a better place to start for safety considerations.

Inference of Mathematical Structure

$\mathcal{C}$ set of classifiers for elements of $\mathcal{D}$

$\mathcal{I} : \mathcal{D} \rightarrow \mathcal{C}$ represents the ideal process we wish to learn

$H = \{f : \mathcal{D} \rightarrow \mathcal{C} \mid \Phi(f)\}$ possible “guesses”

$$\begin{aligned} H_X &= \{f \in H \mid \forall x \in X, f(x) = \mathcal{I}(x)\} \\ &= \{f : \mathcal{D} \rightarrow \mathcal{C} \mid \Phi(f) \wedge \forall x \in X, f(x) = \mathcal{I}(x)\} \end{aligned}$$

We note that no algorithm can do better than choose at random from H_X because all known information of $\mathcal{I}$ is accounted for with Φ and X .

Probably Approximately Correct (PAC) learning

$100 - \alpha\%$ probability of $\beta\%$ or less failure rate.

*We are 90% certain
that the function will only fail in 5% of cases.*

How safe does this make us feel?

Where does it need improvement?

Computational Learning Theory

This does not begin to answer:

- How severe is the failure in that 5%?
- What happens in the 10% where we are wrong?

PAC learning theory focuses on when an algorithm can be trained in polynomial time.

PAC sacrifices these questions for concerns about computational complexity.

Safety Through Characterization

Safe Learning Problems

If our intention is to show that the system is safe, what do we need to ask?

Is the system going to kill me?

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If our intention is to show that the system is safe, what do we need to ask?

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To get a non-probabilistic answer to this, we must fully characterize the behavior of the system.

To do so, we might ask:

- How often is the system wrong? (Frequency)

- In what cases do errors occur? (Dimensionality)

- When it is wrong, how bad are the mistakes? (Severity)

- What kind of errors occur? (Modality)

- What and how much information is needed to “train” the system to an acceptable level of safety? (Complexity)

Errors: Frequency

If $\mathcal{D}$ is a finite set, this is a simple question to answer.
We just need to count cases.

If $\mathcal{D}$ is not finite, we need a measure on $\mathcal{D}$.

In practice, these are computational problems and should be finite.

Care must be taken if we choose to use a measure or we arrive at some of the same safety difficulties we would have had we used probabilistic methods.

Errors: Dimensionality

Consider the problem of self-driving cars.

Errors that occur during interstate driving are probably of more concern than errors during parking.

If we can partition $\mathcal{D}$ (our input data) in such a way that we isolate cases such as parking and interstate driving, then we can simplify the problem by treating the restriction of $\mathcal{I}$ to each partition as a separate system.

Errors: Severity

The most obvious way to do this is to apply a metric to $\mathcal{C}$, but there are situations where errors may not be metric.

Identity – counter example: small errors may have no impact

Symmetry – counter example: overshooting can be more serious than undershooting

Sub-additivity – counter example: two small errors might have less impact than one large error

Non-negativity / Real-valued – counter example: incomparable types of errors

Errors: Modality

The type of error may also be of interest.

There may be two (or more) distinct modes of failure that are not directly comparable.

In full generality, error would be represented as a partially ordered set.

However, we will assume that when these errors are comparable when viewed in terms of safety – hence such a partial order can be embedded into a total order.

Errors: Complexity

Complexity is dependent upon other safety properties.

We can not determine what data is needed without determining dimensionality and modality.

We can not determine how much data is needed without making judgements concerning frequency and severity.

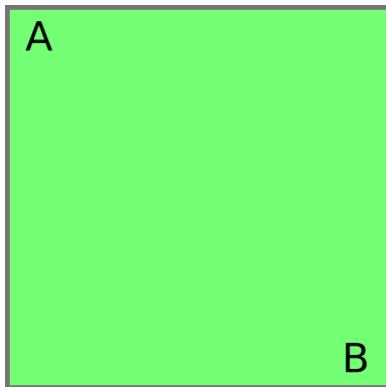
Safety Summary

Using these five safety concepts, we will attempt to analyze an example learning problem.

- Modality – In what ways can the system fail?
- Severity – What errors are unacceptable?
- Dimensionality – Is there a way to partition the input domain to create two or more smaller problems?
- Frequency – How often will the system fail?
- Complexity – What and how much information is needed for safe operation?

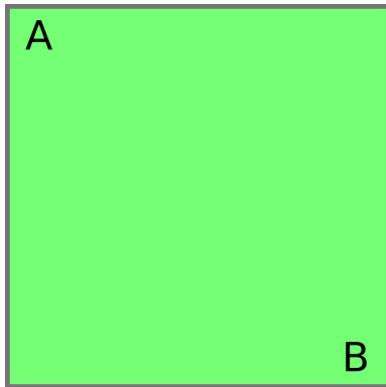
An Example Analysis

Avoid the Cliff



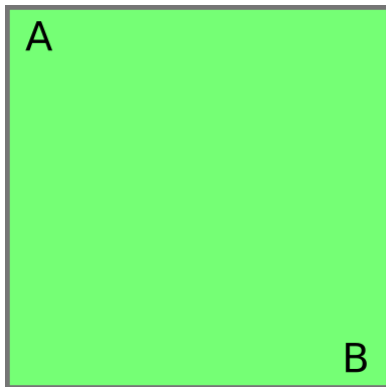
Imagine we have an autonomous robot navigating from some point A to some point B on a bounded flat plane $D \subset \mathbb{R}^2$ with access to sensor input in the form of a data vector from $\mathbb{R}^n$.

Avoid the Cliff



For simplicity, assume that the environment is static and sensors are omnidirectional – that is, sensor input is independent of time, speed, and direction.

Avoid the Cliff



Thus, for every point in D , we have a fixed (but not necessarily unique) input vector. This simplifies the input space we need to consider.

Avoid the Cliff

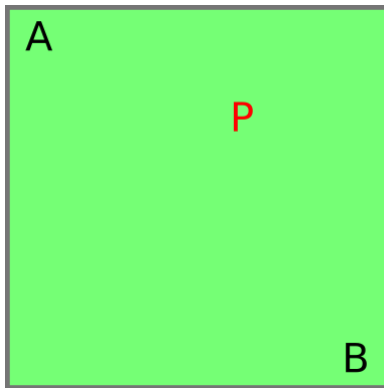
MODALITY – The system may fail in two ways:

- The robot may fail to find it's way to B .
- The robot may fall afoul of some hazard that it can not escape.

SEVERITY – With respect to these two modes of failure:

- The first type is recoverable and represents zero safety risk since operation may continue.
- The second type represents unacceptable safety risk that must be avoided.

Avoid the Cliff



Suppose now that we identify a single safety hazard, such as a cliff, that the robot can not safely traverse at some P in D .

Avoid the Cliff

We ask: Does there exist a set of examples X such that H_X only includes functions which generate paths avoiding P ?

In general, only if we sample a complete boundary around P .

So we consider modifying our hypothesis space H (through Φ) by other methods.

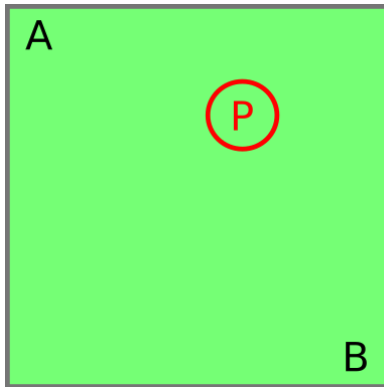
Avoid the Cliff

DIMENSIONALITY – If we can separate the input space into clear regions based on severity and modality, the problem becomes easier.

Since each point in D is not guaranteed to have a unique sensor input vector, this may require additional sensor information. In practice, this could be location data and geofencing or some method of detecting an artificially placed boundary.

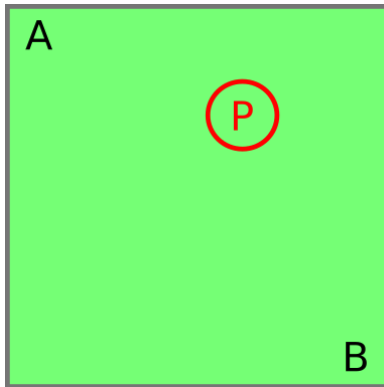
Doing so, we can isolate P in an “unsafe” region.

Avoid the Cliff



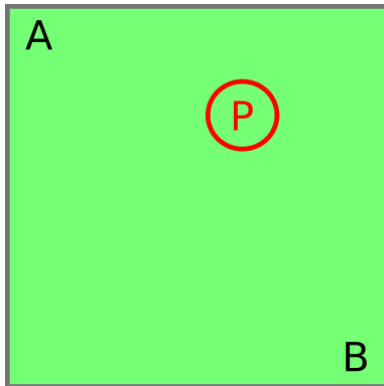
Thus at any point outside of this unsafe region, the robot can freely determine where to go next in a safe manner. However, within the boundary of the unsafe region, the risk is serious.

Avoid the Cliff



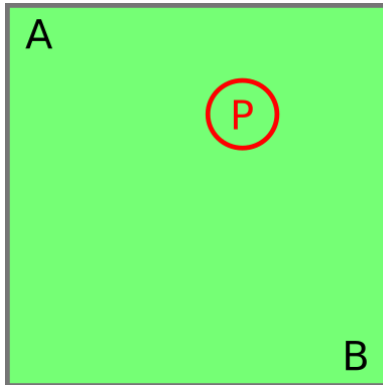
Since the path is continuous, this gives us two cases – the robot is in the clear, or the robot is at the boundary of the unsafe region.

Avoid the Cliff



In the safe area, any direction is safe. At the boundary, safety can be ensured by either forcing the robot to turn back or cease autonomous operation and await instruction.

Avoid the Cliff



The latter case can be handled and verified using run-time assurance techniques that will ensure the robot does not encounter P .

Avoid the Cliff

FREQUENCY – We note that this is a computational problem. Sensor readings can only be evaluated at discrete time intervals. The problem can be discretized so that we are not actually working in a subset of $\mathbb{R}^2$, but on a “checkerboard.”

As long as we can ensure that the robot can not move more than one space without checking sensors, we can be certain that it never passes the boundary.

This gives us a frequency of zero for safety errors.

Avoid the Cliff

COMPLEXITY – Given a zero frequency of safety errors, the complexity is almost trivial. No data is needed to safely train the robot to get from A to B beyond what it takes to establish the boundary.

The robot may never reach its goal, but even pointless meandering carries no safety risk.

Analysis

If (1) we can provide some form of identifiable boundary around P and (2) verify that the boundary is not crossed, then we have a fully safe system.

Alternately, we might use a cautious approach. Since the space is finite we could use sensor data around P to create a no-go area. Because sensor data is not necessarily unique to location, this might lead to avoidance behavior elsewhere.

Of particular note, if there is no way to mark or sample the area around P , there is no way to avoid the hazard. This is understandable because P would be essentially invisible.

Conclusion

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This project did not go in the direction I wanted it to go.

I feel like I am just now at the point where I could consider beginning the actual project I envisioned.

My take-away – justifying mathematical assumptions is difficult when dealing with an applied problem.

Conclusion

- This safety perspective provides the type of framework I was initially searching for at the beginning of this project.
- There is no claim about whether a problem can be solved with algorithmic learning – this perspective only addresses the extent and magnitude of failures.
- It should be further developed and formalized. After further work, it should be applied to a formal safety verification of classical machine learning problems.
- Prove that machine learning problems can not, in general, be bounded for safety.
- Find classes of problems where error can always be bounded and investigate their properties.

Thank You